

RANYA RAJESH PRASAD

Software Engineering · Backend & Full-Stack · AI Systems & Evaluation · Cloud & CI/CD · Published Machine-Learning Research

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RECOGNITION

Corresponding author, peer-reviewed international publication, “SkinSight: An App Leveraging Neural Networks for Skin Cancer Detection,” *Intl. Journal of Pharma Medicine & Biological Sciences* (2025) · **Oral presentation, ICBCB 2025** — IEEE co-sponsored, Seoul National University; submissions from 21 countries · **First Place, shsHacks (judged by Hudson River Trading)** · Illinois State Scholar (top 10%) · IJAS Gold (×2)

EDUCATION

University of Illinois Urbana-Champaign — B.S. Electrical & Computer Engineering, Grainger College of Engineering **Expected May 2029**

Sophomore, **junior standing by credit hours** · GPA 3.65/4.00 · **Minor: Hoeft Technology & Management Program** — joint program of the Gies College of Business and the Grainger College of Engineering, admitted by application as a first-year

Coursework: ECE 220 Computer Systems & Programming (C, C++, assembly, memory), CS 173 (Discrete Structures), MATH 241 (Multivariable Calculus), MATH 257 (Linear Algebra) · **CS 225 Data Structures (Spring 2027)** — **self-studying ahead of enrollment**: AVL trees, heaps, hash tables, graphs in C++.

Data structures & algorithms applied in projects: suffix-array (Burrows-Wheeler) alignment and external sorting of multi-GB files (genome pipeline) · dependency-tree traversal and hash-set keyword matching (NLP study) · database indexing, constraints and locking under concurrency (Cafe Queues)

TECHNICAL SKILLS

Languages: Python, Java, C++, C, TypeScript, JavaScript, SQL · data structures & algorithms, object-oriented design

Backend & Data: REST API design (Flask, Node.js / Express), **PostgreSQL** (normalized schemas, constraints, transaction locking), MySQL, Firebase, message protocols, concurrency · **Frontend & Mobile**: React, HTML / CSS, Android (Java)

Cloud & DevOps: AWS (Cognito, ECS/Fargate, ECR, CloudWatch), Terraform, Docker, GitHub Actions CI/CD, pytest, Gradle, Git, code review, Linux

AI & ML: LLM systems (Claude API, MCP servers, evaluation harnesses), TensorFlow / Keras, TensorFlow Lite, Transfer Learning, NLP (spaCy, NLTK, VADER), statistical testing · **extensive use of Claude Code and GitHub Copilot**, with review and tests on everything they write

EXPERIENCE

Jr. Applied AI Developer (IRAP) — AbbVie · Information Research Academic Partnerships **Oct 2026 – Present**

– Building **evaluation frameworks that decide when AI-generated output can be trusted** — golden test sets, regression detection, **latency and cost dashboards** — wired into CI/CD with automated gates and rollback.

Software Developer — NOBE · client engagement for a registered investment advisory firm (paying client) · Python, React **Sep 2026 – Present**

– **Backend for financial software on brokerage trade data**: extending the firm's Investment Engine — a Python backend and React front end that **ingests trade data across brokerage platforms and runs automated cost-basis analysis** — to the client's requirements, so advisors can run several strategies in one account.

– **Data model**: building a sub-account model where **holdings, cost basis and ledger history are scoped per sub-account**, so a strategy can be opened, transferred between, reintegrated or closed **without touching the others**.

– **Performance-measurement engine**: implementing time-weighted rate of return (TWRR) per sub-account, where contributions, withdrawals, dividends and — the hardest case — **transfers of holdings between sub-accounts** all distort the numbers. **Validated against a client-supplied 13-scenario workbook that serves as the answer key and reference test suite** for exactly these edge cases.

– **Roll-up reporting**: sub-account allocations combine into one account view **weighted by dollar value, not averaged percentages**, and decompose back down.

– **Delivered the initial portfolio navigation workflow** of the advisor interface for client review.

Software Developer Intern — Fylm TV, Inc. · platform infrastructure: AWS, Terraform, Docker, CI/CD · distributed team **Aug 2026 – Present**

– Own the Terraform module provisioning the platform's authentication layer on AWS (Cognito) in a shared codebase; hardened the **pipeline gating every pull request** (validate, test, secret scanning) so CI never mutates live state.

Software Chair — WinnieWheels autonomous delivery robot · Women in ECE (WECE), UIUC · C++, Node.js, React **Oct 2025 – Present**

– **Appointed as a first-year student**, one of two software chairs **leading software for a 50-person team**: owned the architecture and **ran code reviews**.

– **Designed a real-time, three-tier system**: the timing-critical obstacle-avoidance loop runs in C++ firmware on the microcontroller, a control server coordinates, and the client never talks to hardware directly, so the robot stays safe if the network stalls.

– **Specified the messaging protocol** between firmware and server (newline-delimited JSON over USB serial) and built the bridge that parses sensor frames and writes commands back; **found and fixed a saturated link** by throttling and deduplicating commands.

– Built a **simulator mode** on the identical code path so the full stack ran with no hardware; ran live at Engineering Open House, driven by hundreds of visitors.

PROJECTS

SkinSight — skin cancer detection app · Java, Android, TensorFlow Lite · Lead & Corresponding Author **2023 – 2025**

– **Compared four neural-network approaches over three trials each** — ANN (31.19%), FFNN (64.20%), CNN (73.58%) and **Transfer Learning (84.69% mean, 90.32% best)**, a statistically significant gain over the CNN (t-test, $p = 0.021$) — then shipped the best model in a **Java Android app that runs it on the device** through TensorFlow Lite. In 2026 upgraded the build with GitHub Copilot (Java 17, Gradle 7.6.4).

Cafe Queues — user-facing order-queue backend · PostgreSQL, Flask (Python), React · cafe-queues.vercel.app **2022 – 2026**

– Campus pre-order site students used daily at peak lunch. Added a **Flask REST backend (four endpoints) on a normalized PostgreSQL schema**, with the no-duplicate-ticket invariant **enforced in the database by a UNIQUE constraint and a transaction lock rather than in application code** — **verified under concurrency: 20 simultaneous orders issue 20 distinct tickets**.

NetWatch — AI log-triage agent where the evaluation is the merge gate · Python, Claude API, GitHub Actions, pytest **2026**

– Parses device syslog and detects link flaps, BGP adjacency loss and authentication storms with z-score baselines plus rules. **Instrumented every model call for latency, tokens and cost**; CI fails on any accuracy drop or >50% cost / p95 latency growth and re-verifies the previous version, so rollback is tested.

Distributed parallel computing — genome variant-calling pipeline · supercomputer nodes (Linux, Google Cloud) · UChicago **2023 – 2024**

– **Parallel computing where runtime is the cost**: a head node dispatched each job to worker nodes that aligned, genotyped and annotated in parallel, tuned by worker count and threads per node; **identified alignment and annotation as the bottleneck**. Scaled from one chromosome (2 hours, \$18.75) to all 23 (16 hours, \$150) — **23× the chromosomes at 8× the cost**, against days for other methods; 148,361 high-quality variants.

Decoding Bias — data-ingestion and sentiment pipeline · Python, spaCy, VADER · UCSB Summer Research Academies, published co-author **2024**

– **Owned keyword engineering and pipeline optimization** for a two-stage Python scraper over the 50 most-viewed news outlets (14,000+ articles); **designed the 71-keyword lexicon that set the sample frame**. Sentences classified by spaCy dependency parsing and VADER sentiment; chi-squared test, $p < .05$.

EARLIER WORK & LEADERSHIP

Crisis Management App (2022) — Java Android app on a Firebase database shared across users' devices; integrated the Contacts, Location, SMS, Camera and Google Maps APIs. **weather-alert-mcp** (2026) — one API contract implemented in Python and TypeScript; malformed calls rejected at the boundary. **Forsys Inc.**, Data Analytics Intern (2022) — SQL, Tableau dashboards, Twitter sentiment analysis. **Leadership**: EXPO Technical Lead & EOH Special Events Director, Engineering Council · Project Technology Lead, Startups in Business & Technology Club.